
Decode-Position Channel Drift at Long-Decode Reasoning: Budget Tuning as a Partial Cross-Model Remedy

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Abstract

Quamba2 argues that SSM activations exhibit channel-order preservation and activation persistence, properties that support offline per-channel quantization for Mamba-family models. We test the corresponding assumption at reasoning-scale decode horizons by measuring decode-position channel-set drift: whether channels in the top 1% by magnitude early in decode remain in the same protected set later. Across Granite-4.0-H-Small, Nemotron-3-Nano, DeepSeek-R1-Distill-Qwen-1.5B, and Falcon-H1, strict top-1% set-leaving is 0.566234756098, 0.533713200380, 0.670572916667, and 0.673611111111, respectively, at decode horizons up to 20,000 tokens. Layer-type stratification shows comparable drift at block outputs for SSM/Mamba, attention, and MoE blocks, so the Quamba2-style persistence assumption fails at the measured hybrid-model block-output surface rather than only in a single component type. We then test W4A16 channel-protection policies with frozen controls. Hard position or scale switches are actively harmful: M2 and M10 are beaten by random-bin controls. EMA smoothing removes that random-control-beats failure but is insufficient at top-1% budget: M11 recovers only 0.048299284138 median of the BF16-vs-static gap. Cross-tensor activation+K coupling has signal relative to random controls but not enough recovery at top-1%. Budget tuning is the first partially positive result. M11b top-5 recovers 0.449284091125 on Granite-Small, but with wide CI95 [-1.300900018791, 1.000790626547]; on Nemotron, top-5 recovers 0.456736183270 with CI95 [0.345887792388, 0.794986745913] but does not beat static top-10, while top-10 recovers 0.814739798903 and beats static top-10 by 0.220356055286 median. Paro-Quant recovers 0.753776848891 on Granite-Small via rotation, showing that the problem is not W4A16 impossibility but channel-set protection fragility. A per-position KL analysis finds sublinear rather than superlinear growth, with AR decays 0.4387–0.5147, weakening compound-error explanations in this packet. The resulting contribution is a scoped mechanism paper with a partial budget-tuning remedy, not a settled deployable quantization recipe or a runtime/memory efficiency implementation.

1 Introduction

Quamba2 argues that SSM activations have channel-order preservation and activation persistence, properties that motivate offline per-channel quantization and channel clustering for Mamba1/Mamba2 backbones (Chiang et al., 2025). We show that the corresponding protected-set assumption breaks down at reasoning-scale decode horizons in our W4A16 setting: across three hybrid reasoning LLMs (Granite-4.0-H-Small, Nemotron-3-Nano, Falcon-H1) and one pure-Transformer reference (DeepSeek-R1-Distill-Qwen-1.5B), 53–67% of top-1% magnitude channels at calibration positions strictly leave the top-1% set by the final measured decode position. This is not merely within-set rank shuffling; it is strict set membership change. The distinction matters for deployment. Static protected-channel maps, rotations, per-channel formats, and calibration-time outlier policies are easiest to engineer when the protected set remains stable. If channels leave the protected set as decode position increases, then a policy calibrated early in a trace can be stale later in the same trace.

The deployment pressure is not abstract. Artificial Analysis benchmark-cost data reported by Bort (2025) put OpenAI o1 at \$2,767.05 to evaluate across seven reasoning-heavy benchmarks, compared with \$108.85 for GPT-4o on the same suite; the report attributes the roughly $25\times$ cost gap to both token expansion and higher per-token prices. Gartner’s March 2026 inference-cost forecast predicts that trillion-parameter LLM inference will become over 90% cheaper by 2030 than in 2025, while warning that advanced reasoning demand can still make total compute scarce (Gartner, 2026). Commercial market forecasts are not evidence about model internals, but they indicate scale: one MarketsandMarkets report projects the AI inference market from \$106.15B in 2025 to \$254.98B in 2030, and another projects edge AI hardware from \$26.14B in 2025 to \$58.90B in 2030 (MarketsandMarkets, 2025a;b). Long-decode quantization failures therefore matter both for datacenter reasoning workloads and for eventual edge reasoning deployments.

The scope here is deliberately narrow. For large-scale standard-architecture reasoning models, recent work such as ParoQuant reports substantial W4A16 quality recovery (Liang et al., 2025). The frontier tested in this draft is different: small reasoning models, hybrid Mamba-2 or parallel-hybrid architectures, pure PTQ rather than distillation or quantization-aware training, and long-decode W4A16 channel-level protection.

The current status is a scoped measurement and mechanism paper with one partial remedy. The empirical result is stronger than the original Granite/Mamba-specific framing: decode-position channel drift appears in Granite hybrid models, a Nemotron-3 hybrid, a pure-Transformer DeepSeek control, and Falcon-H1’s parallel hybrid topology. The negative method result is also sharper: static unions, hard position bins, hard scale bins, top-1% EMA smoothing, top-1% cross-tensor coupling, DecDEC-style reactive selection, and stable-core-only protection do not produce a reliable W4A16 channel-set method. The partial positive result is budget tuning. M11b shows that larger EMA-smoothed protected budgets can recover a meaningful fraction of the BF16-vs-static gap, but the best budget differs by model and matched static top-10 protection is already strong on Nemotron. The paper therefore claims that budget is load-bearing, not that a fixed top-5 EMA recipe is a settled cross-model method.

The contribution is:

- quantitative characterization of decode-position channel-set drift at reasoning-scale horizons (up to 20,000 decode tokens) across hybrid Mamba-2 and pure-Transformer reasoning LLMs, extending DecDEC’s short-horizon Transformer evidence to the long-decode reasoning regime while contradicting the channel-persistence assumption that motivates Quamba2-style static ordering;
- cross-architecture replication on Nemotron-3, DeepSeek-R1-Distill, and Falcon-H1 under fixed deterministic traces;
- strict set-membership decomposition showing that many channels leave the protected top-1% set entirely, rather than merely shuffling within it;
- mechanism evidence that boundary-discontinuous switching is harmful, top-1% smoothing has signal but insufficient magnitude, cross-tensor coupling has signal without enough budget, and per-position KL does not show superlinear compound-error growth in the measured Granite packet;
- a partial remedy: M11b budget-tuned EMA protection recovers 0.4493 median on Granite-Small at top-5 and 0.8147 median on Nemotron at top-10, but the Nemotron top-5 arm does not beat matched static top-10, so the method is a budget-sensitivity finding rather than a model-invariant recipe;
- a defensive SOTA context: ParoQuant’s algorithmic Granite baseline recovers 0.7538 median through rotation, showing that the paper’s negative results are scoped to channel-set protection rather than to all W4A16 PTQ.

The paper does not claim a deployable kernel/runtime system, a universal quantization recipe, or completed architecture-general validation. In particular, M11b is reported as evidence about the protected-budget mechanism; we do not implement or benchmark a production M11b runtime.

2 Related Work

Mamba quantization and dynamic outliers. OutlierMigrate is not the first work to observe dynamic outlier behavior in Mamba-style models. In vision state-space models, QMamba reports that hidden state sequences exhibit highly dynamic variations and introduces Temporal Group Quantization for hidden states (Li et al., 2025). OuroMamba makes the point even more directly for vision Mamba models: it identifies dynamic activation-outlier variation across time steps as a failure mode for static post-training quantization and uses lightweight dynamic outlier detection during inference (Ramachandran et al., 2025). These results are important prior art. They establish that dynamic outliers are already known in the vision-Mamba setting.¹

Language-Mamba quantization work has emphasized a different operating regime. Mamba-PTQ shows that recurrent LLMs exhibit activation outlier channels and frames outlier-aware quantization as a first step (Pierro and Abreu, 2024). Quamba proposes a static 8-bit per-tensor recipe that suppresses maximum input values to selective SSMS and uses a Hadamard transform for output activations (Chiang et al., 2024). Quamba2 is closer to the assumption tested here: it explicitly builds on channel-order preservation and activation persistence for Mamba1/Mamba2 quantization (Chiang et al., 2025). Our contribution is not that dynamic channel sets exist in every setting; it is that the Quamba2-style persistence assumption does not hold for the measured reasoning-scale W4A16 decode horizons. MambaQuant uses variance-aligned rotations and reports heavy-tailed, channel-uneven distributions in Mamba-family models (Xu et al., 2025). Quamba-SE softens the treatment of activation outliers with multiple adaptive scales but still evaluates a per-value quantizer rather than a decode-time channel migration policy (Chen and Hemani, 2026). OutlierMigrate asks which regime applies to hybrid LLMs in long reasoning traces: the static or calibration-time protection regime common in language deployment recipes, or the dynamic regime already documented in vision Mamba.

Static protection in LLM deployment. Several influential LLM quantization systems protect or transform salient activation structure using calibration-time statistics or fixed transformations. SmoothQuant migrates activation difficulty into weights using offline activation statistics (Xiao et al., 2023); AWQ protects a small set of activation-salient weight channels (Lin et al., 2023); QuaRot removes hidden state outliers with rotations (Ashkboos et al., 2024); KVQuant isolates KV cache outliers per vector and uses per-channel key quantization (Hooper et al., 2024); and BlockDialect assigns block-wise number formats from a learned formatbook (Jang and Tambe, 2025). These systems are not claimed to be wrong by our current evidence. The implication is narrower: if the protected high-magnitude set changes substantially during long decode in hybrid LLMs, static protection maps may need migration-aware updates or validation on the specific recurrent/linear-attention architecture. Several of these systems already include adaptive components, such as KVQuant’s per-vector dense-and-sparse handling or BlockDialect’s online activation format selection; our current evidence motivates stress-testing those designs on hybrid long decode, not declaring them failed.

Dynamic channel and KV-cache policies. Dynamic inference-time policies are closer to the methods we test and fail. DecDEC dynamically selects salient activation channels at each decode step and fetches residual compensation from CPU memory (Park et al., 2025). DecDEC also reports low recall for static outlier-channel identification at short decode horizons. Our measurement is therefore not a first demonstration that channel sets change during decode; the defensible distinction is horizon, model class, architecture, and benchmark surface: up to 20,000 decode tokens on reasoning LLMs, including hybrid Mamba-2 and parallel-hybrid models, rather than DecDEC’s short-horizon Transformer setting. M2 and hard-binned

¹The phrase “outlier migration” is overloaded. In this paper, the operational measurement is decode-position channel drift: which activation channels are in the top-magnitude set at different decode positions. SmoothQuant uses migration for activation-to-weight difficulty transfer (Xiao et al., 2023). MoBiQuant uses it for token-set shifts under different bit widths (Wang et al., 2026); we reserve our quantitative claims for the channel-across-decode-position phenomenon measured in this paper.

M10 differ from DecDEC in that they switch a precomputed protected-channel or scale policy by decode-position bin; M11 tests whether smoothing that decision surface avoids the boundary discontinuities exposed by M2 and M10. It does avoid the random-control-beats pattern but still does not recover enough quality. KV-cache systems address a different tensor. KIVI quantizes key cache per channel and value cache per token (Liu et al., 2024); PM-KVQ targets long-CoT KV-cache quantization and RoPE-induced key-cache distribution shift (Liu et al., 2025); and AttentionPredictor learns to predict future attention scores for KV-cache compression and prefetch (Yang et al., 2025). Newer long-context KV-cache systems sharpen the distinction. ChanMix reallocates bits across sensitive KV-cache channels (Liao and Wen, 2026), while MixKVQ uses query-aware mixed precision for critical key channels in long-context reasoning (Zhang et al., 2025). These are close neighbors on the channel-wise long-context quantization axis, but they act on cached key/value tensors. Our measurements and interventions target linear-layer activation channels, so a ChanMix- or MixKVQ-style success would not by itself resolve the W4A16 activation-channel failure measured here. These works motivate long-decode inference adaptation, but our measurements are about linear-layer activation channel sets.² PMPD changes precision across inference phases and deeper decode positions (Chen et al., 2025); our current evidence points in an orthogonal direction, where later decode positions can require different protected channels rather than simply lower precision.³

Quantization error accumulation. Statistically-Lossless Quantization (SLQ) argues that, in a near-lossless compression regime, quantization error need not compound across token positions (Helcig et al., 2026). That counter-result is important for this draft because one possible explanation for the failed W4A16 interventions is long-horizon compound error rather than channel selection alone. The regimes are not the same: SLQ studies statistically lossless compression, whereas this draft studies aggressive W4A16 PTQ, small reasoning models, and endpoints up to 20,000 decode positions. QEP identifies layer-wise error propagation as a limitation of low-bit PTQ (Arai and Ichikawa, 2025), and LAQuant argues that long-decoding reasoning failures can persist despite short-decode accuracy unless calibration and lookahead losses account for downstream residual effects (Choi et al., 2026). Activation Sensitivity gives the theoretical version of the same warning: activation magnitude alone is only a proxy for loss sensitivity, and cross-layer error accumulation, calibration mismatch, and task-conditional sensitivity are open challenges for PTQ (Xu, 2026). We therefore treat compound error as an open hypothesis, not a conclusion. Our KL-accumulation packet directly tests this mechanism in the aggressive W4A16 endpoint setting and finds flat or sublinear trajectories rather than superlinear growth, so the paper does not use compound error as the primary explanation for the observed channel-set failures.

Channel-wise gated recurrence. Recent efficient-recurrence and linear-attention architectures make data-dependent channel-wise memory control a central design axis. Gated Linear Attention (GLA) formulates linear attention as recurrent matrix-valued state with data-dependent gating (Yang et al., 2023). Gated DeltaNet combines adaptive memory erasure with delta-rule updates and reports hybrids that mix Gated DeltaNet with attention or Mamba2 layers (Yang et al., 2024). Moonshot AI’s October 2025 Kimi Linear report is a recent example in this line: its Kimi Delta Attention (KDA) is described as channel-wise gated linear attention that extends Gated DeltaNet with finer-grained gating, and the released Kimi Linear model is a layerwise hybrid of KDA and Multi-Head Latent Attention (Moonshot AI Kimi Team, 2025). The official Qwen3.6 model card likewise describes a hybrid layout that interleaves Gated DeltaNet blocks with gated attention (Qwen Team, 2026). These sources motivate OutlierMigrate as an empirical test of whether high-magnitude channels stay fixed during long decode in hybrid language models. They do not imply that Kimi Linear, Qwen3.6, GLA, or other unmeasured models exhibit the migration measured in this draft.

²“Hot channels” in HCP/CHON (arXiv:2602.02047) refers to persistently extreme channels during NVFP4 pretraining, which is close in vocabulary but opposite in setting from the drifting W4A16 inference-time channels measured here.

³We refer to arXiv:2603.19296 as test-time TTQ when discussing it, to avoid confusion with classical Trained Ternary Quantization.

3 Method

Phase 0 follows the frozen preregistration at `experimental/outlier_migrate/phase0/preregister_om_phase0.md`. The model is `ibm-granite/granite-4.0-h-tiny` at HuggingFace snapshot commit `791e0d3d28c86e106c9b6e0b4cecddee0375b6124`. The prompt set contains 12 deterministically selected AIME-2025 traces, indices 0–11, with prompt SHA `sha256:2f27c54baa8448e033d6e82f53f775dc6abe38188e4f1e5c0b97e3c74fe7c1dd` from `experimental/outlier_migrate/phase0/results/om_phase0_20260508T011824Z/prompt_manifest.json`.

Phase 1 follows the frozen preregistration at `experimental/outlier_migrate/phase1/preregister_om_phase1.md`. The model is `ibm-granite/granite-4.0-h-small` at HuggingFace snapshot commit `b8c0982bab7fde4eb48110f5a069527c008fab39`. The prompt set contains 24 deterministically selected AIME-2025 traces, indices 0–23, with prompt SHA `sha256:aa038b29332b6d137d558205ee441163e7ea4cb3cc323eb705a2f5928fd2fe4e` from `experimental/outlier_migrate/phase1/results/om_phase1_20260508T014959Z/prompt_manifest.json`.

For every transformer layer, the runner records per-channel activation magnitudes. Phase 0 records decode positions `{100, 500, 1000, 5000, 10000}`; Phase 1 records `{100, 500, 1000, 5000, 10000, 20000}`. Each gate selects the top 1% channels by mean magnitude at position 100 and computes the fraction whose rank at the final preregistered position differs by more than two channel positions. The reported gate metric is the trace-bootstrap aggregate over layers with equal layer weighting.

3.1 Comparison with DecDEC measurement

DecDEC is the closest prior empirical measurement of decode-time channel-set change. Its Figure 5 evaluates the recall of static analysis-based outlier identification against a per-step oracle top-K channel set in short Transformer decode. Our measurement differs in four ways. First, the horizon is reasoning-scale: up to 20,000 decode positions rather than roughly 100 decode steps. Second, the model class is reasoning LLMs rather than chat-tuned Transformers. Third, the architecture set includes hybrid Mamba-2 and parallel-hybrid models, with DeepSeek-R1-Distill-Qwen-1.5B as a pure-Transformer reference. Fourth, the prompts are AIME-style long reasoning traces rather than WikiText/BBH-style evaluation.

The numerical values should not be compared as if they were the same metric. DecDEC reports low recall for static outlier identification; our main decomposition reports strict set-leaving from the position-100 top-1% set to the final measured top-1% set, plus within-set rank shuffling. Both measurements are consistent with a shared warning: static channel identities can become stale during decode. Our contribution is the long-horizon reasoning-scale extension and the associated W4A16 method-failure analysis, not the first observation that outlier-channel sets can change.

4 Experiments

The Phase 0 result packet is `experimental/outlier_migrate/phase0/results/om_phase0_20260508T011824Z`. The checker output is `experimental/outlier_migrate/phase0/results/om_phase0_20260508T011824Z/checker_result.json`, the metrics file is `experimental/outlier_migrate/phase0/results/om_phase0_20260508T011824Z/metrics.json`, and artifact completeness is recorded in `experimental/outlier_migrate/phase0/results/om_phase0_20260508T011824Z/artifact_check.json`. The activation artifact SHA is `sha256:2783aa329d82e8f43ea4f342e52f0a25fc283b0b8971880f2da093739474d687` from `experimental/outlier_migrate/phase0/results/om_phase0_20260508T011824Z/activation_magnitude_manifest.json`.

The Phase 1 result packet is `experimental/outlier_migrate/phase1/results/om_phase1_20260508T014959Z`. The checker output is `experimental/outlier_migrate/phase1/results/om_phase1_20260508T014959Z/checker_result.json`, the metrics file is `experimental/outlier_migrate/phase1/results/om_phase1_20260508T014959Z/metrics.json`,

and artifact completeness is recorded in `experimental/outlier_migrate/phase1/results/om_phase1_20260508T014959Z/artifact_check.json`. The activation artifact SHA is `sha256:326a04351e0dda28cf919fe4745d7d9341011db13d940a06cf9c8470633003e1` from `experimental/outlier_migrate/phase1/results/om_phase1_20260508T014959Z/activation_magnitude_manifest.json`.

The partial Phase 2 result packet is `experimental/outlier_migrate/phase2/results/om_phase2_nemotron3_20260508T231723Z`. The checker output is `experimental/outlier_migrate/phase2/results/om_phase2_nemotron3_20260508T231723Z/checker_result.json`, the metrics file is `experimental/outlier_migrate/phase2/results/om_phase2_nemotron3_20260508T231723Z/metrics.json`, and artifact completeness is recorded in `experimental/outlier_migrate/phase2/results/om_phase2_nemotron3_20260508T231723Z/artifact_check.json`. The model is `nvidia/NVIDIA-Nemotron-3-Nano-30B-A3B-BF16` at snapshot commit `cbd3fa9f933d55ef16a84236559f4ee2a0526848`. The activation artifact SHA is `sha256:bc841b938a125f05649f84577c0ddb7ab287213a683892cb81ea6ca06296206d` from `experimental/outlier_migrate/phase2/results/om_phase2_nemotron3_20260508T231723Z/activation_magnitude_manifest.json`. This packet is explicitly scoped as partial Nemotron-3-only validation: `Qwen3.6` and `Kimi Linear` are deferred, and `full_cross_validation_complete=false`.

The Phase 3 intervention packet is `experimental/outlier_migrate/phase3/results/om_phase3_20260509T212000Z`. It follows the frozen preregistration at `experimental/outlier_migrate/phase3/preregister_om_phase3_intervention.md`. The checker output is `experimental/outlier_migrate/phase3/results/om_phase3_20260509T212000Z/checker_result.json`, the metrics file is `experimental/outlier_migrate/phase3/results/om_phase3_20260509T212000Z/metrics.json`, the per-trace perplexities and recoveries are in `experimental/outlier_migrate/phase3/results/om_phase3_20260509T212000Z/per_trace_metrics.json`, and artifact completeness is recorded in `experimental/outlier_migrate/phase3/results/om_phase3_20260509T212000Z/artifact_check.json`. The model is again `ibm-granite/granite-4.0-h-tiny` at snapshot commit `791e0d3d28c86e106c9b6e0b4cecddee0375b6124`. The prompt set uses the fixed 24-trace Phase 1 set with prompt SHA `sha256:aa038b29332b6d137d558205ee441163e7ea4cb3cc323eb705a2f5928fd2fe4e`. The quantization scheme is simple symmetric per-channel INT4 weights with FP16 activations and BF16 protected-channel rows; no AWQ-style or SmoothQuant-style activation-aware scaling is used. The scored window is the 64-token window ending at decode position 10000. The preregistered recovery metric is

$$1 - \frac{\text{perplexity}_{\text{regime}} - \text{perplexity}_{\text{BF16}}}{\text{perplexity}_{\text{static1\%}} - \text{perplexity}_{\text{BF16}}}.$$

The pass rule is median recovery ≥ 0.50 with bootstrap 95% CI lower bound > 0.30 . The kill rule is recovery < 0.20 or CI upper < 0.30 . The mandatory controls are static position-100 top-2% protection and magnitude-average top-1% protection over positions $\{100, 1000, 5000, 10000\}$.

The preregistered dynamic-outlier pass rule in both gates requires migration fraction at least 0.05 and bootstrap CI lower bound greater than 0.05. This is a nonstationarity screen, not a claimed practically meaningful effect-size standard. Phase 1 uses the same dynamic path because Phase 0 returned `PASS_OM_PHASE0_DECODE_TIME_MIGRATION`.

5 Results

The original preregistered migration fraction intentionally conflates two effects: channels can leave the top-1% set entirely, or they can remain in the set while shuffling rank by more than two positions. For static channel protection, the stricter set-membership readout is the more direct post-hoc interpretability measure, not the preregistered gate criterion. A channel is counted as strict set-leaving only if it is in the per-trace, per-layer top-1% set at decode position 100 and has final rank greater than the top-1% boundary at decode position 10000 in Phase 0 or 20000 in Phase 1. Under that definition, 0.634244791667 of Phase 0 base

Quantity	Phase 0	Phase 1
Decision	dynamic-outlier pass	replicated-at-scale pass
Strict set-leaving fraction	0.634244791667	0.566234756098
Strict set-leaving 95% CI	[0.605208333333, 0.664192708333]	[0.550076219512, 0.581707317073]
Within-set rank-shuffling fraction	0.175260416667	0.270934959350
Within-set rank-shuffling 95% CI	[0.160156250000, 0.191015625000]	[0.260797764228, 0.280614837398]
Original preregistered migration fraction	0.8178385416666667	0.843165650406504
Original migration 95% CI	[0.797265625, 0.8368489583333334]	[0.8334349593495936, 0.8511432926829268]
Preregistered threshold	≥ 0.05 and CI lower > 0.05	≥ 0.05 and CI lower > 0.05
Artifact complete	true	true
Model	ibm-granite/granite-4.0-h-tiny	ibm-granite/granite-4.0-h-sma
		ll
Trace count	12	24
Layer count / hidden size	40 / 1536	40 / 4096
Decode positions	100, 500, 1000, 5000, 10000	100, 500, 1000, 5000, 10000, 20000
Bootstrap seed	20260508	20260508

Table 1: OutlierMigrate Phase 0 and Phase 1 gate results. Phase 0 values are from experimental/outlier_migrate/phase0/results/om_phase0_20260508T011824Z/checker_result.json and experimental/outlier_migrate/phase0/results/om_phase0_20260508T011824Z/metrics.json; strict decomposition values are from experimental/outlier_migrate/phase0/results/om_phase0_20260508T011824Z/migration_decomposition.md. Phase 1 values are from experimental/outlier_migrate/phase1/results/om_phase1_20260508T014959Z/checker_result.json and experimental/outlier_migrate/phase1/results/om_phase1_20260508T014959Z/metrics.json; strict decomposition values are from experimental/outlier_migrate/phase1/results/om_phase1_20260508T014959Z/migration_decomposition.md.

top-1% channels and 0.566234756098 of Phase 1 base top-1% channels leave the protected set entirely. The within-set shuffling rates are 0.175260416667 and 0.270934959350, respectively. The gate decision remains based only on the original preregistered migration metric.

The exact Phase 0 provenance values are: model snapshot commit 791e0d3d28c86e106c9b6e0b4cecddee0375b6124, prompt SHA sha256:2f27c54baa8448e033d6e82f53f775dc6abe38188e4f1e5c0b97e3c74fe7c1dd, and activation artifact SHA sha256:2783aa329d82e8f43ea4f342e52f0a25fc283b0b8971880f2da093739474d687; these are recorded in experimental/outlier_migrate/phase0/results/om_phase0_20260508T011824Z/model_provenance.json, experimental/outlier_migrate/phase0/results/om_phase0_20260508T011824Z/prompt_manifest.json, and experimental/outlier_migrate/phase0/results/om_phase0_20260508T011824Z/activation_magnitude_manifest.json. The exact Phase 1 provenance values are: model snapshot commit b8c0982bab7fde4eb48110f5a069527c008fab39, prompt SHA sha256:aa038b29332b6d137d558205ee441163e7ea4cb3cc323eb705a2f5928fd2fe4e, and activation artifact SHA sha256:326a04351e0dda28cf919fe4745d7d9341011db13d940a06cf9c8470633003e1; these are recorded in experimental/outlier_migrate/phase1/results/om_phase1_20260508T014959Z/model_provenance.json, experimental/outlier_migrate/phase1/results/om_phase1_20260508T014959Z/prompt_manifest.json, and experimental/outlier_migrate/phase1/results/om_phase1_20260508T014959Z/activation_magnitude_manifest.json.

The Phase 0 checker returned PASS_OM_PHASE0_DECODE_TIME_MIGRATION. Its exact artifact reason was that the point estimate 0.81783854 is at least 0.05 and the CI lower bound 0.79726562 is greater than 0.05. The Phase 1 checker returned PASS_OM_PHASE1_REPLICATED_AT_SCALE. Its exact artifact reason was that the

Quantity	Partial Phase 2 Nemotron-3
Decision	partial PASS; Qwen3.6/Kimi deferred
Strict set-leaving fraction	0.533713200380
Strict set-leaving 95% CI	[0.467414529915, 0.599982193732]
Within-set rank-shuffling fraction	0.269082383666
Within-set rank-shuffling 95% CI	[0.238841405508, 0.299056267806]
Original preregistered migration fraction	0.820809591642925
Original migration 95% CI	[0.7865325261158594, 0.8544931149097815]
Preregistered threshold	≥ 0.05 and CI lower > 0.05
Artifact complete	true
Model	nvidia/NVIDIA-Nemotron-3-Nano-30B -A3B-BF16
Trace count	24
Layer count / hidden size	52 / 2688
Decode positions	100, 500, 1000, 5000, 10000, 20000

Table 2: Partial Phase 2 cross-family validation. The checker-backed packet is experimental/outlier_migrate/phase2/results/om_phase2_nemotron3_20260508T231723Z. This is not full cross-validation because Qwen3.6 and Kimi Linear are deferred.

Run	Original drift	Strict set-leaving	Within-set shuffling	Packet
Granite-Small Phase 1	0.836763211382	0.566234756698	0.270934959350	experimental/outlier_migrate/phase9/step9_0_decomposition_replication.md
Nemotron-3 Phase 2	0.801667853751	0.533713200380	0.269082383666	experimental/outlier_migrate/phase9/step9_0_decomposition_replication.md
DeepSeek-R1-Distill Phase 5'	0.834914434524	0.670572916667	0.165736607143	experimental/outlier_migrate/phase9/step9_0_decomposition_replication.md
Falcon-H1 Phase 7	0.767887205387	0.673611111111	0.097537878788	experimental/outlier_migrate/phase7/results/om_phase7_falcon_h1_20260512T223600Z/checker_result.json

Table 3: Cross-architecture decomposition after Phase 5' and Phase 7. Strict set-leaving remains above 0.50 on Granite-Small, Nemotron-3, DeepSeek, and Falcon-H1. The Falcon-H1 packet uses post-block residual hooks; it is not pathway-specific pre-sum attribution.

point estimate 0.84316565 is at least 0.05 and the CI lower bound 0.83343496 is greater than 0.05.

Phase 5' broadens the phenomenon beyond measured Mamba-2-style hybrids: deepseek-ai/DeepSeek-R1-Distill-Qwen-1.5B returns DYNAMIC_REGIME_TRANSFORMER with original drift 0.8393787202380952, 95% CI [0.8278459821428571, 0.8507254464285714], strict set-leaving 0.670572916667, and within-set shuffling 0.165736607143. Phase 7 checks Falcon-H1 as a parallel hybrid topology and returns WITHIN_LINEAGE_2_CONSISTENT: original drift 0.7833543771043772, 95% CI [0.7544191919191919, 0.8112373737373737], strict set-leaving 0.673611111111, and within-set shuffling 0.097537878788. These two additions make the Mamba-specific title untenable and motivate the broader “decode-position channel drift” framing.

Experiment D formalizes the decomposition and Experiment E checks threshold sensitivity. The top-1% decomposition is stable through top-2% on all landed packets. Top-5% changes the component split and is treated as sensitivity, not the primary operating point. Phase 9 Step 9.0 additionally computes adjacent-bin Jaccard overlap for top-channel sets; the overall top-1% adjacent-bin Jaccard is 0.745999647119, which made position-binned methods plausible enough to test.

5.1 Static and position-conditioned protection attempts

Phase 3 tests the most direct intervention suggested by the measurement: instead of protecting only the position-100 top-1% channels, protect the union of top-1% channels observed at decode positions {100, 1000, 5000, 10000}. This is intentionally a static-set intervention. It asks whether membership migration alone explains enough of the quantization gap to support a simple deployment recipe.

The intervention kills. The checker returned the decision string KILL_OM_PHASE3_INTERVENTION_FAILS because the union regime had median recovery 0.000000000000, below the preregistered 0.20 kill floor. It also missed the pass

Regime	Mean perplexity	Median perplexity	Median recovery	Recovery 95% CI
BF16 baseline	1.005358821083984	1.000282908378305	reference	reference
static-1% position 100	1.015228246226743	1.000282342709987	reference	reference
static-2% position 100	1.021467448964073	1.000390886047853	0.000000000000	[0.000000000000, 0.356017001423]
magnitude average top-1%	1.009248547916613	1.000236584155715	0.061276118929	[0.000000000000, 0.512245438462]
migration-aware union	1.008573512406127	1.000275313560728	0.000000000000	[0.000000000000, 0.711143244199]

Table 4: Phase 3 intervention and controls on Granite-4.0-H-Tiny. Values are from experimental/outlier_migrate/phase3/results/om_phase3_20260509T212000Z/per_trace_metrics.json, experimental/outlier_migrate/phase3/results/om_phase3_20260509T212000Z/metrics.json, and experimental/outlier_migrate/phase3/results/om_phase3_20260509T212000Z/control_metrics.json. Recovery is defined relative to the BF16-vs-static-1% gap.

Grid	Positions	Median recovery	Recovery 95% CI
sparse	{100, 5000, 10000}	0.000000000000	[0.000000000000, 0.752281234025]
primary	{100, 1000, 5000, 10000}	0.000000000000	[0.000000000000, 0.711143244199]
dense	{100, 500, 1000, 2000, 5000, 7500, 10000}	0.000000000000	[0.000000000000, 0.920007799726]

Table 5: Position-grid sensitivity. Values are from experimental/outlier_migrate/phase3/results/om_phase3_20260509T212000Z/grid_sensitivity.md and experimental/outlier_migrate/phase3/results/om_phase3_20260509T212000Z/grid_sensitivity_metrics.json. Increasing grid density did not produce a nonzero median recovery.

rule: median recovery ≥ 0.50 with CI lower bound > 0.30 . The artifact packet is complete, and the control-stop condition did not fire. The best control, magnitude-average protection, exceeded union by 0.061276118929 median recovery, below the human stop threshold of 0.10. Ten of the 24 traces had no recoverable static gap under this scoring definition. That fraction, 0.416666666667, also exceeds the preregistered 25% no-gap kill condition. The checker reports the first kill reason it reaches, but the packet satisfies both kill paths.

The recovery statistic is intentionally centered on the per-trace BF16-vs-static-1% gap, so it can be negative when a regime is worse than static-1%, greater than one when a regime beats BF16 on the scored window, and zero when there is no recoverable static gap. In the union regime, 11 traces had positive recovery, 10 were zero because there was no recoverable static gap, 3 were negative, and 4 exceeded recovery 1.0. This unstable per-trace surface is part of the negative result rather than a hidden success case: the primary statistic was preregistered as the median across all 24 traces. The median was chosen for the gate because the recovery ratio can become very large when the static gap is small; a mean recovery would be dominated by unstable high-ratio traces rather than by the typical trace.

Phase 4 repeats the union intervention on Granite-4.0-H-Small with a larger 512-token scoring window. It also kills: median recovery is 0.000000000000, CI95 [0.000000000000, 0.06954064195470389], and the no-recoverable-static-gap fraction is 0.375. This rules out the simplest explanation that Phase 3 failed only because Granite-Tiny was too robust or because the 64-token scoring window was too short.

Phase 9 M2 then tests a lightly position-conditioned intervention. Instead of one static union, M2 uses three protected sets: early positions {100, 200, 500}, middle positions {1000, 2000, 5000}, and late positions {7000, 10000, 15000}. The active protected set switches at decode positions 800 and 6000. The result kills by the random-control rule. On the vacation-mode 12-trace deterministic slice, 8 traces have a positive static-1% gap. M2 median recovery is -0.866837313391, static-3% matched-cost median recovery is -0.625927638959, and random-bin median recovery is -0.199289023223. The random-bin assignment beats M2 by 0.667548290168 median recovery, exceeding the preregistered 0.10 kill threshold. The random-bin control is not good in absolute terms; its importance is that it is less bad than the intended position-conditioned assignment.

M10 tests whether the problem is not membership but scale choice: it stores separate SmoothQuant-style scale tables for decode-position bins. This also kills by the random-control rule. M10 median recovery is 0.234447927834, CI95 [-2.072905653258708, 0.5035214944499808], while the random-bin scale control has median

Attempt	Model	Decision	Median recovery	CI95	Key control result
Phase 3 union	Granite-Tiny	KILL_OM_PHASE3_INTERVENTION_FAILS	0.000000000000	[0.000000000000, 0.711143241199]	10/24 no recoverable static gap
Phase 4 union	Granite-Small	KILL_OM_PHASE4_INTERVENTION_FAILS	0.000000000000	[0.000000000000, 0.060540641953]	9/24 no recoverable static gap
Phase 9 M2 bias	Granite-Small	KILL_M2_RANDOM_CONTROL_BEATS	-0.86083731391	[-3.435289251335, 0.595238665766]	random-bin median beats M2 by 0.667548290168
Phase 9 M10 scales	Granite-Small	KILL_M10_RANDOM_CONTROL_BEATS	0.23447927834	[-2.072905653259, 0.503521494450]	random-bin median beats M10 by 0.761487459046
Phase 9 M11 EMA	Granite-Small	KILL_M11_AMBIGUOUS	0.048299284138	[-8.941156151014, 0.704599255873]	random-walk median is lower by 1.25440832497
Phase 9 M18 act+K	Granite-Small	KILL_M18_AMBIGUOUS	-0.343590844126	[-14.0771191710979, 0.740692124280]	KIVI key-only median -2.174128311393; random-coupled median -4.664379731051
Phase 9 DecDEC proxy	Granite-Small	PASS_DECDEC_BASELINE_REPORTED	-0.070034781258	[-8.495693501233, 0.674176370350]	random reactive median -1.262604926125; static top-10% median -0.228273308210
Phase 9 M11b top-5	Granite-Small	PASS_M11B_BUDGET_MATTERS	0.449284091125	[-1.300900018791, 1.000799626547]	beats static top-10% by 0.501051267680 median
Phase 9 M11b top-5	Nemotron-3	AMBIGUOUS_M11B_NEMOTRON_BUDGET_SIGNAL	0.456736183270	[0.345887792388, 0.794986745913]	does not beat static top-10%; margin -0.137647560348
Phase 9 M11b top-10	Nemotron-3	PASS_M11B_NEMOTRON_REPLICATES	0.814739789803	[0.254439288178, 0.922552264880]	beats static top-10% by 0.220560555286 median
Phase 9 M26 stable core	Granite-Small	AMBIGUOUS_M26	0.177615807648	[-0.116448558060, 0.999915031324]	beats random matched-core by 1.727084589454 median
Phase 9 ParoQuant	Granite-Small	PASS_PAROQUANT_BASELINE_REPORTED	0.753776848891	[0.477044452405, 1.003769554323]	rotation baseline beats M11b top-5 by 0.304492757767 median

Table 6: Intervention attempts. Phase 3 and Phase 4 test static union protection under W4A16. Phase 9 M2 tests position-conditioned protected sets A/B/C with a static-3% matched-cost control and a random-bin negative control. Phase 9 M10 tests hard-binned scale tables. Phase 9 M11 tests EMA-smoothed protected-set updates. Phase 9 M18 tests joint activation and key-cache channel protection. The DecDEC row is a descriptive algorithmic baseline, not a positive-method gate. M11b is the only channel-set method with a partial positive signal, but the best budget differs by model and Nemotron top-5 does not beat matched static top-10. ParoQuant is a rotation baseline, not a channel-set method. Checker decision strings are audit labels; the paper framing is determined by the control comparisons in the final column. M10 values are from experimental/outlier_migrate/phase9/resu lts/om_phase9_m10_granite_small_vac12_20260515T085800Z/checker_result.json; M11 values are from experimental/outlier_migrate/phase9/results/om_phase9_m11_granite_small_vac12_20260516T010728Z/checker_result.json; M18 val- ues are from experimental/outlier_migrate/phase9/results/om_phase9_m18_gr anite_small_vac12_20260516T193500Z/checker_result.json; DecDEC values are from experimental/outlier_migrate/phase9/results/om_phase9_decdec_granite _small_vac12_20260517T141500Z/checker_result.json; M11b Granite values are from experimental/outlier_migrate/phase9/results/om_phase9_m11b_granite_s mall_vac12_reuse_20260518T030300Z/checker_result.json; M11b Nemotron val- ues are from experimental/outlier_migrate/phase9/results/om_phase9_m11b_n emotron_static1pct_salvage_20260524T2130Z/checker_result.json; M26 values are from experimental/outlier_migrate/phase9/results/om_phase9_m26_granit e_small_vac12_20260518T203000Z/checker_result.json; ParoQuant values are from experimental/outlier_migrate/phase9/results/om_paroquant_granite_small_20260520T 1555Z/checker_result.json.

recovery 0.995935386880 and beats M10 by 0.761487459046. M11 then removes the hard boundary by replacing bin switches with an EMA-smoothed protected-set score. M11 does not pass: the best arm, alpha 0.5, has median recovery 0.048299284138 with CI95 [-8.94115615101423, 0.7045992558730417]. However, its random-walk control has median recovery -1.206105548349, so M11 is structurally different from M2 and M10: smoothing removes the random-control-beats failure, but smoothing alone does not recover quality.

Together, the intervention results support a three-mechanism account. First, boundary dis- continuities are actively harmful: M2 and M10 are both beaten by random-bin controls. Second, smoothing removes that harm but does not solve the gap at top-1% budget: M11 is not beaten by random-walk, yet misses the positive-method bar. Third, signal quality and budget separate. M18 activation+K has median recovery -0.343590844126, while KIVI key-only and random coupled controls have medians -2.174128311393 and -4.664379731051. That comparison suggests the coupled selection is not arbitrary noise, but it does not re- cover quality at top-1%. DecDEC then tests a reactive endpoint-selection baseline: median recovery is -0.070034781258 with CI95 [-8.495693501233, 0.674176370350]. It is less dam- aging than a random reactive top-1% control whose median is -1.262604926125, but it does not recover the W4A16 static gap. M11b changes the picture by increasing budget under a smoothed update. Granite-Small top-5 recovers 0.449284091125 median and beats static top-10 by 0.501051267680 median, though the CI is wide. Nemotron top-5 is also positive with a strictly positive CI, but static top-10 is stronger; Nemotron top-10 is the arm that beats static top-10. The conclusion is therefore not "EMA top-5 works" but "budget is load-bearing, and the budget-selection rule is not yet model-invariant."

A post-M18 diagnostic pass further narrows what the current packets can say. The K/V- cache migration question is not identifiable from existing artifacts: only M18 contains K/V- channel evidence, and that evidence is endpoint-only at decode position 10000 rather than

early-to-late K/V snapshots. The recovery curve question is also not identifiable, because M2, M10, M11, and M18 each score one 512-token endpoint window. The activation-side evidence is still useful: Granite-Small fine-grid top-1% Jaccard changes smoothly across 500-position bins, and all measured models retain a small always-protected core that is smaller than the full top-1% set. These diagnostics make DecDEC and M11b the right next tests rather than another unmotivated method. Those packets have now landed: DecDEC tests reactive endpoint selection, while M11b tests whether the remaining failure is simply protection-budget insufficiency.

5.2 Budget tuning, rotation context, and KL accumulation

M11b directly tests the budget-insufficiency hypothesis by running the M11 EMA policy with larger protected-channel budgets. On Granite-Small, the top-5 arm is the first mechanical channel-set PASS: median recovery is 0.449284091125, and it beats the static top-10 control by 0.501051267680 median. The uncertainty is still large, with CI95 [-1.300900018791, 1.000790626547], so the paper treats it as a signal requiring cross-model validation rather than as a settled method.

The Nemotron Path C salvage supplies that validation surface after rerunning the corrected static_1pct baseline. It resolves the previous infrastructure failure: static_1pct now records 6052 quantized tensors and 76 excluded tensors. The result is positive but not a clean top-5 transfer. Top-5 median recovery is 0.456736183270 with CI95 [0.345887792388, 0.794986745913], but static top-10 recovers 0.594383743618. Top-10 M11b recovers 0.814739798903 with CI95 [0.254439288178, 0.922552264880] and beats static top-10 by 0.220356055286 median. This supports the budget mechanism while showing that the best budget and matched-static comparison are model-dependent.

ParoQuant provides a different context. Our algorithmic ParoQuant baseline on Granite-Small recovers 0.753776848891 with CI95 [0.477044452405, 1.003769554323], beating M11b top-5 by 0.304492757767 median. This is not a contradiction of the channel-drift measurement. It shows that a rotation baseline can reduce the same W4A16 gap through a different mechanism, so the paper’s negative claims must be scoped to channel-set protection rather than to W4A16 PTQ generally.

Finally, the KL-accumulation packet weakens the compound-error explanation in this specific Granite-Small setting. It computes per-position $KL(BF16||Q)$ over 44,208 rows with dense-grid fallback. Mean KL is 0.150302750276 for static-1%, 0.132701884446 for the DecDEC proxy, and 0.131406585383 for M11. The best fits are flat or sublinear rather than superlinear, and the fitted AR decay parameters are 0.5147 for static-1%, 0.4461 for the DecDEC proxy, and 0.4387 for M11. Thus the present evidence does not support runaway long-horizon compounding as the main failure mode; signal staleness and budget selection remain the stronger explanations.

5.3 Layer-stratified migration

The negative intervention result does not erase the measurement finding: it narrows the mechanism. Figure 1 uses the existing Phase 0, Phase 1, and partial Phase 2 activation packets to stratify strict set-leaving and within-set shuffling by layer type. The result is not confined to a single Granite mixer family. Granite shows strict set-leaving in both attention and SSM/Mamba layers, while Nemotron-3 shows it in attention, SSM/Mamba, and MoE-classified layers. This is consistent with the paper’s diagnostic claim: static membership is unstable in the measured packets, but successful handling likely needs more than a larger static protected set.

6 Discussion

The strongest claim in this draft is now broader and more precise than the initial Granite-specific hypothesis. Across four measured model families, channels that are top-1% by magnitude early in a long decode often do not remain in the later top-1% set. The strict

Run / layer type	Layers	Strict set-leaving	Within-set shuffling
Phase 0 Granite-Tiny attention	4	0.618490	0.186198
Phase 0 Granite-Tiny SSM/Mamba	36	0.635995	0.174045
Phase 1 Granite-Small attention	4	0.557165	0.282266
Phase 1 Granite-Small SSM/Mamba	36	0.567243	0.269676
Phase 2 Nemotron-3 attention	6	0.563014	0.254630
Phase 2 Nemotron-3 MoE	23	0.526503	0.277778
Phase 2 Nemotron-3 SSM/Mamba	23	0.533280	0.264157

Figure 1: Layer-stratified migration decomposition from experimental/outlier_migrate/p_hase3/results/layer_stratified_migration.json. Values report layer-type means, averaged equally over layers. Strict set-leaving is the portion most directly relevant to static protected-channel membership; within-set shuffling motivates scale-refresh mechanisms but is not itself a membership failure.

set-leaving decomposition is the most important systems readout because it directly targets the membership assumption behind static protected-channel maps. The original rank-drift metric remains the preregistered gate; the stricter decomposition explains why the gate matters for deployment.

The method results are informative because they separate mechanisms rather than collapsing every failure into one explanation. Static union protection addresses only membership; it does not refresh scales, respond to within-set rank changes, or decide which layers need different treatment. Phase 4 shows that the Phase 3 failure is not just a Granite-Tiny or 64-token window artifact. M2 and M10 show that hard protected-set or scale-table binning can be actively harmful. M11 then separates smoothness from recovery: it removes the random-control failure but does not restore quality at top-1%. M18 and DecDEC show that reactive or cross-tensor signals are better than matched random controls but still insufficient at top-1%. M26 shows that the always-protected stable core has signal but insufficient magnitude. M11b shows that budget can unlock recovery, while Nemotron shows that the budget optimum does not simply transfer.

The KL result rules out one tempting explanation in the measured packet. If compound error were the dominant mechanism, we would expect per-position KL to grow superlinearly or maintain very high error autocorrelation. Instead, the trajectories are flat or sublinear and the AR decays are about 0.44–0.51. This is consistent with the SLQ counter-result in spirit, though in a much more aggressive W4A16 regime. The paper therefore emphasizes stale or misbudgeted channel protection rather than runaway accumulation.

ParoQuant constrains the negative claim. A rotation baseline recovers 0.753776848891 median on Granite-Small, which is better than M11b on the same packet. The paper should therefore be read as evidence that channel-set protection is fragile in long-decode W4A16 reasoning, not as evidence that W4A16 PTQ is impossible. The natural ICLR follow-up is not another threshold-tuned channel set, but composition of rotation-based conditioning with budget-aware protection, plus broader replication at larger reasoning model scales.

The same result has a long-window state-management interpretation. A static protected-channel map is a small internal state summary: it says which channels deserve high precision later in the generated sequence. Strict set-leaving means that this summary becomes stale as the decode window grows. From that perspective, M2/M10 are failed abrupt state refresh policies, M11 is a smoother but underbudgeted state refresh policy, DecDEC is a reactive per-step refresh baseline, and M11b shows that the state summary needs enough capacity to be useful. This is why the result is relevant beyond a single quantizer: long-context systems that compress, cache, or internalize hidden state need to know whether their summary variables remain valid across thousands of generated tokens.

The theoretical mechanism remains plausible but is not itself an empirical claim about unmeasured models. In channel-wise gated recurrence or linear-attention layers, independent per-channel decay rates imply that channels retain state at different rates. Low-decay channels can accumulate magnitude over long decode, while high-decay channels shed magnitude. Under heterogeneous inputs and heterogeneous decay rates, the composition of the top 1% channels by magnitude should change across decode positions rather than remain fixed to the position-100 set. Kimi Linear’s KDA, Qwen3.6-style Gated DeltaNet blocks, and GLA-style gated recurrence provide architectural examples where channel-wise or fine-grained data-dependent gating is central (Moonshot AI Kimi Team, 2025; Qwen Team, 2026; Yang et al., 2024; 2023). This draft has not measured Kimi Linear, Qwen3.6, or GLA variants.

7 Limitations

This draft is not camera-ready final and should not be read as a deployable systems method. It reports replicated measurement, failed method classes, and a partial budget-tuning remedy, not an end-to-end efficiency implementation. The strongest positive evidence is the cross-model measurement plus the M11b budget-sensitivity result; the strongest negative evidence is that many channel-set policies fail or depend strongly on model and budget.

The scope is W4A16 post-training quantization for small reasoning models and hybrid Mamba-2 or parallel-hybrid architectures, with a small pure-Transformer reference. It should not be read as an all-formats claim. The underlying decode-position channel-set drift may be a broader autoregressive phenomenon, but quality consequences depend on format dynamic range: INT4 exposes severe failure surfaces here, FP8 may absorb comparable drift in settings like ThoughtFlow, block formats such as MXFP4/NVFP4 move the analysis from channels to blocks, and INT2/INT3 would likely be still more fragile. Large standard reasoning models are also outside the current evidence. Recent reasoning PTQ work such as ParoQuant reports strong W4A16 recovery on larger reasoning models (Liang et al., 2025). Our algorithmic ParoQuant Granite-Small baseline also recovers a large fraction of the BF16-vs-static gap, so the negative result must stay scoped to channel-set protection methods rather than to W4A16 PTQ in general.

The empirical surface is still limited. All reported traces are deterministic AIME-2025 slices. The bootstrap uncertainty is over traces, not over independent benchmark families, seeds, or prompt distributions. The top-1% fraction, rank-delta threshold, and decode-position grid were fixed in the packets; they should not be retuned on these results. Experiment E shows reasonable top-1% to top-2% stability, but top-5% changes the component split. Several recovery packets have fewer effective traces than the nominal 12 because recovery is undefined when the corrected static baseline has no recoverable BF16-vs-static gap. Granite M11b uses 8 positive-gap traces, while Nemotron M11b uses 10; the CIs should be read with that small effective sample size in mind. The Phase 9 methods were preregistered or audit-authorized as mechanism tests, not selected post hoc as a winner search.

Cross-lineage validation remains incomplete. Nemotron-3, DeepSeek, and Falcon-H1 reduce Granite-specific risk, but Qwen3.6 was blocked by hook accessibility in current SGLang infrastructure without source modification, and Kimi Linear has not been measured. Falcon-H1 is post-block residual evidence rather than separate pre-sum SSM/Attention pathway attribution. Generalization to RWKV-7, GLA-style systems, and other efficient-recurrence variants is not established.

Finally, the method failures do not prove that decode-position channel drift is unusable. They prove that static union sets, a larger Granite-Small scoring window, hard-binned policies, top-1% EMA smoothing, top-1% activation/KV coupling, DecDEC-style reactive selection, and stable-core-only protection are not enough under this quantization setup. M11b shows that budget tuning can matter, but Nemotron shifts the strongest arm from top-5 to top-10 and makes the matched static top-10 control competitive. The next credible evidence should come from rotation-plus-budget composition, larger-model replication, and training-time interventions rather than threshold retuning on the same failed surfaces.

The KL result is also scoped. It is a Granite-Small analysis over static-1%, DecDEC proxy, and M11 regimes, not a proof that compound error cannot appear under other quantizers, budgets, models, or decoding distributions.

8 Reproducibility Statement

The Phase 0 result packet is `experimental/outlier_migrate/phase0/results/om_phase0_20260508T011824Z`. It includes `environment.json`, `model_provenance.json`, `prompt_manifest.json`, `command_metadata.json`, `random_seed.json`, `metrics.json`, `bootstrap_ci.json`, `checker_result.json`, and `artifact_check.json`. The recorded Phase 0 command was:

```
experimental/shared/run_phase0_branch.py --branch outlier_migrate
```

The Phase 1 result packet is `experimental/outlier_migrate/phase1/results/om_phase1_20260508T014959Z`. It includes the same artifact classes and records the Granite-4.0-H-Small snapshot commit `b8c0982bab7fde4eb48110f5a069527c008fab39`. The recorded Phase 1 command was:

```
experimental/outlier_migrate/phase1/run_om_phase1.py
```

The checker decisions and exact numbers in this draft are taken from the two Granite result directories above and the partial Phase 2 directory `experimental/outlier_migrate/phase2/results/om_phase2_nemotron3_20260508T231723Z`. The OutlierMigrate runners emit the artifact-checked metric file `metrics.json`; they do not emit a separate `profiler_metrics.json`. The recorded Phase 2 command was:

```
experimental/outlier_migrate/phase2/run_cross_model.py \
--partial-nemotron3-only \
--run-id om_phase2_nemotron3_20260508T231723Z \
--batch-size 4 --dtype bfloat16 --device auto
```

The Phase 3 intervention packet is `experimental/outlier_migrate/phase3/results/om_phase3_20260509T212000Z`. It includes `environment.json`, `model_provenance.json`, `prompt_manifest.json`, `command_metadata.json`, `random_seed.json`, `per_trace_metrics.json`, `metrics.json`, `control_metrics.json`, `grid_sensitivity_metrics.json`, `checker_result.json`, and `artifact_check.json`. The recorded Phase 3 command was:

```
experimental/outlier_migrate/phase3/run_phase3_intervention.py \
--run-id om_phase3_20260509T212000Z \
--batch-size 4 --dtype bfloat16 --device auto \
--reuse-prequant-run-dir \
experimental/outlier_migrate/phase3/results/om_phase3_20260509T180200Z
```

The grid-sensitivity artifact was generated with:

```
experimental/outlier_migrate/phase3/run_grid_sensitivity.py \
--run-dir \
experimental/outlier_migrate/phase3/results/om_phase3_20260509T212000Z
```

Figure 1 is a LaTeX-native summary table generated from `experimental/outlier_migrate/phase3/results/layer_stratified_migration.json`.

The Phase 4 intervention packet is `experimental/outlier_migrate/phase4/results/om_phase4_20260511T054000Z`. The Phase 5' Transformer-control packet is `experimental/outlier_migrate/phase5_prime/results/om_phase5p_20260512T053800Z`. Experiment D and E artifacts are in `experimental/outlier_migrate/decomposition_analysis/`. The Phase 7 Falcon-H1 packet is `experimental/outlier_migrate/phase7/`

results/om_phase7_falcon_h1_20260512T223600Z. Phase 9 Step 9.0 outputs are experimental/outlier_migrate/phase9/step9_0_decomposition_replication.md and experimental/outlier_migrate/phase9/step9_0_bin_overlap_analysis.md. The Phase 9 M2 packet is experimental/outlier_migrate/phase9/results/om_phase9_m2_granite_small_vac12_finalized_20260514T233800Z. Its checker decision is KILL_M2_RANDOM_CONTROL_BEATS; the vacation-mode adaptation notes are in swarm/vacation_decisions/. The Phase 9 M10 packet is experimental/outlier_migrate/phase9/results/om_phase9_m10_granite_small_vac12_20260515T085800Z; the Phase 9 M11 packet is experimental/outlier_migrate/phase9/results/om_phase9_m11_granite_small_vac12_20260516T010728Z; and the Phase 9 M18 packet is experimental/outlier_migrate/phase9/results/om_phase9_m18_granite_small_vac12_20260516T193500Z. Their checker decisions are KILL_M10_RANDOM_CONTROL_BEATS, KILL_M11_AMBIGUOUS, and KILL_M18_AMBIGUOUS. Post-M18 analysis reports are in experimental/outlier_migrate/phase9/post_m18_analysis/; they document which mechanism questions are identifiable from the landed packets and which would require new telemetry.

The DecDEC algorithmic baseline packet is experimental/outlier_migrate/phase9/results/om_phase9_decdec_granite_small_vac12_20260517T141500Z. The M11b Granite packet is experimental/outlier_migrate/phase9/results/om_phase9_m11b_granite_small_vac12_reuse_20260518T030300Z. The M26 stable-core packet is experimental/outlier_migrate/phase9/results/om_phase9_m26_granite_small_vac12_20260518T203000Z. The KL accumulation packet is experimental/outlier_migrate/phase9/results/om_phase9_kl_granite_small_dense_20260519T085400Z. The ParoQuant algorithmic baseline packet is experimental/outlier_migrate/phase9/results/om_paroquant_granite_small_20260520T1555Z. The valid Nemotron M11b Path C salvage packet is experimental/outlier_migrate/phase9/results/om_phase9_m11b_nemotron_static1pct_salvage_20260524T2130Z. The original Nemotron packet is not used for recovery claims because its static_1pct baseline was invalid; the Path C packet reruns that baseline with corrected W4A16 quantization and reuses only validated score caches.

9 Ethics Statement

This is an activation-analysis measurement study on benchmark prompts. The current artifact does not make deployment, user-data, safety, or systems-efficiency claims. The main ethical risk is overclaiming the replicated measurement or the partial M11b budget signal as a validated deployable method. This draft therefore separates measurement, mechanism, baseline context, and partial-remedy claims explicitly.

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